

# **VISVESVARAYA TECHNOLOGICAL UNIVERSITY**

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**Project Report**

**on**

**GA And ANN based Heart Disease Prediction System**

Submitted in partial fulfillment of the requirements for the VII semester

**Bachelor of Engineering**

**in**

**Computer Science and Engineering**

**of**

Visvesvaraya Technological University, Belagavi.

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**CAMBRIDGE INSTITUTE OF TECHNOLOGY**

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**BANGALORE - 560 036**

**2025-2026**

**CAMBRIDGE INSTITUTE OF TECHNOLOGY**  
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**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**



**CERTIFICATE**

Certified that **Mr.Mahammad Zuber, Mr.Mohammed Umar, Mr.Nandan M K** bearing **USN: 1CD23CS406, 1CD23CS409, and 1CD23CS411** respectively, are bonafide students of **Cambridge Institute of Technology**, have successfully completed the project entitled “**GA and ANN based Heart Disease Prediction System**“ in partial fulfillment of the requirements for VII semester **Bachelor of Engineering in Computer Science and Engineering** of **Visvesvaraya Technological University, Belagavi** during academic year 2025-2026. It is certified that all Corrections/Suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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# DECLARATION

We, **Mr. Mahammad Zuber, Mr. Mohammed Umar, Mr. Nandan N K** bearing **USN 1CD23CS406, 1CD23CS409, and 1CD23CS411** respectively, are the students of VII semester, Computer Science and Engineering, Cambridge Institute of Technology, hereby declare that the project entitled **“GA and ANN based Heart Disease Prediction System”** has been carried out by us and submitted in partial fulfillment of the course requirements of VII semester **Bachelor of Engineering in Computer Science and Engineering** as prescribed by **Visvesvaraya Technological University, Belagavi**, during the academic year 2025-2026.

We also declare that, to the best of our knowledge and belief, the work reported here does not form part of any other report on the basis of which a degree or award was conferred on an earlier occasion on this by any other student.

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## ACKNOWLEDGEMENT

We would like to place on record our deep sense of gratitude to **Shri. D. K. Mohan**, Chairman, Cambridge Group of Institutions, Bangalore, India for providing excellent Infrastructure and Academic Environment at CITech without which this work would not have been possible.

We are extremely thankful to **Dr. B V Ravishankar**, Principal/Director, CITech, Bangalore, for providing us the academic ambience and everlasting motivation to carry out this work and shaping our careers.

We express our sincere gratitude to **Dr. Shreekanth M Prabhu**, HOD, Dept. of Computer Science and Engineering, CITech, Bangalore, for his stimulating guidance, continuous encouragement and motivation throughout the course of present work.

We wish to extend our thanks to **Dr Shilpa V**, Project Coordinator, Dept. of CSE, CITech, Bangalore, for her critical, insightful comments, guidance and constructive suggestions to improve the quality of this work

We also wish to extend our thanks to **Prof. Shilpa S B** Assistant Professor Dept. of CSE, CITech for her guidance and impressive technical suggestions to complete our project..

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# **ABSTRACT**

Cardiovascular disease remains one of the leading causes of death worldwide, and urgent and accurate diagnosis remains crucial to arresting further loss of life. This project aim to develop a hybrid Heart Disease Prediction System integrating Genetic Algorithm (GA) and Artificial Neural Network (ANN) methodologies to enhance predictive robustness and accuracy. The system applies GA to select optimal features, thereby reducing data dimensionality and determining the most significant clinical indicators of heart disease. Optimized features are passed as input to an already trained ANN model that learns complex, non-linear data patterns to predict the likelihood of heart disease in patients. Experimental results on standard medical datasets demonstrate that the GA+ANN hybrid model outperforms traditional machine learning approaches with respect to accuracy, precision, and recall. This project seeks to advance Machine Learning based Prediction system for early detection or life threatening disease.

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# CHAPTER 1

## INTRODUCTION

Cardiovascular disease (CVD) is one of the most severe public health problems worldwide and makes a significant contribution to morbidity and mortality in all countries. CVD and its complications impose a great burden on healthcare systems, hospitals, and the economy. Heart diseases are the leading cause of death globally year after year. It is very important to study this project intends to help in the advancement of healthcare technology and developing early preventative measures against heart diseases. The traditional diagnosis of heart disease is based on extensive physical examination, patient history, and numerous diagnostic tests, including electrocardiograms (EKGs), echocardiograms, and blood tests. While these tests are effective in identifying cardiovascular conditions, they are time-consuming and, in the process, tend to overlook early symptoms unconsciously, thereby making early identification more difficult and increasing the likelihood of serious consequences.

### 1.1 Background

Disease prediction has the potential to become a cornerstone of health diagnosis in the future with the proper training of Machine Learning models, We will be able to produce an early diagnosis of disease and formulate preventative measures for the treatment. However, there exists research gaps in currently existing prediction models which we will discuss further during literature review. But first lets us discuss already existing information about heart disease prediction. There has been a previous research that investigated how **Genetic Algorithm (GA)** can be used to **optimize Artificial Neural Networks (ANNs)** for predicting **cardiovascular disease (CVD)**, their research yielded 68.35% accuracy with standalone ANN, And 73.43% accuracy by optimizing the model with Genetic Algorithm. The research also highlighted the improvement in accuracy compared other ANN models. The research only utilized GA for Hyper-parameter Tuning that is a thing to keep in mind.

So, to follow up from that the purpose of our project is to advance the existing research by achieving higher average accuracy and better utilizing Genetic Algorithms for not only Hyper-parameter tuning but also feature selection to optimize the ANN prediction model.

## 1.2 Genetic Algorithms

Genetic Algorithms (GAs) are inspired by natural evolution and work by selecting, recombining, and mutating candidate solutions until the best one emerges. Genetic Algorithms (GAs) should be used in projects involving disease predictions because they provide an efficient and systematic way to optimize Artificial Neural Networks (ANNs), which otherwise rely on trial-and-error for selecting the best layers, neurons, activation functions, and other hyper-parameters. Since medical datasets are large, complex, and highly non-linear, manually tuning ANN parameters often leads to poor accuracy or models getting stuck in local minima. GA overcomes these issues by evolving multiple candidate network configurations, maintaining diversity, and exploring the search space more effectively. This results in a more accurate, robust, and generalizable prediction model. In the context of heart disease prediction, GA-optimized ANN significantly improves accuracy over traditional ANN and other machine-learning models, making it a highly suitable and impactful optimization method for such critical healthcare applications.

Our project will be developed to select most important features from information from medical information. Features are individual pieces of data, like age, blood pressure, or cholesterol. All features do not have the same impact on prediction accuracy. Techniques such as chi-square, ANOVA, and mutual information will be used to select features of most influence. Removing redundant features or redundant features simplifies the model and makes it faster and more robust. We use the Genetic Algorithms to do this feature selection.

# CHAPTER 2

## LITERATURE REVIEW

### 2.1 Introduction

The literature review is conducted to identify and examine the existing research work in the chosen field to gain valuable information. Literature review was performed to understand the existing systems and find a suitable approach for the development of a Heart Diseases prediction model. As the study was made to compare different proposed systems, the literature review was performed to identify the most effective system.

### 2.2 Related Works

**[1] Improved Study of Heart Disease Prediction System Using Data Mining Techniques, Dangare & Apte, 2012.**

This paper investigates the effectiveness of traditional data mining classification techniques for heart disease prediction using an enhanced feature set. In addition to the commonly used 13 clinical attributes, the authors incorporate obesity and smoking as additional risk factors, aiming to improve diagnostic accuracy. The study compares Neural Networks, Decision Trees (J48), and Naïve Bayes classifiers on heart disease datasets from the UCI repository. Experimental results indicate that Neural Networks achieve the highest prediction accuracy (100%), outperforming Decision Trees and Naïve Bayes. The analysis underscores the importance of incorporating relevant lifestyle-related features and demonstrates the superiority of neural network models for complex medical classification tasks. However, the study primarily focuses on accuracy and does not deeply address issues such as overfitting or model interpretability.

**[2] Predicting Alzheimer's Disease Using 3D Convolutional Neural Networks, Payan & Montana, 2013.**

This paper presents a deep learning-based approach for early Alzheimer's disease (AD) diagnosis using structural MRI scans. The authors propose a novel framework combining sparse auto-encoders with 3D Convolutional Neural Networks (3D-CNNs), enabling the model to exploit full volumetric brain information rather than relying on

2D slices. Using the ADNI dataset comprising 2,265 MRI scans across healthy controls, MCI, and AD patients, the study demonstrates that 3D-CNNs significantly outperform traditional machine learning classifiers and 2D CNN approaches. The analysis highlights the advantage of learning spatially localized 3D patterns in brain anatomy, which are crucial for neuro-degenerative disease detection. Overall, the paper establishes 3D deep learning architectures as a powerful and state-of-the-art solution for automated Alzheimer's disease prediction.

**[3] GA–NN Approach for ECG Feature Selection in Rule-Based Arrhythmia Classification, G"ozde Aslanta,s, Fikret G"urgen, Albert Ali Salah, 2014.**

This study proposes a hybrid Genetic Algorithm–Neural Network (GA-NN) framework for arrhythmia detection using ECG signals, with a strong emphasis on feature reduction and interpretability. The genetic algorithm is used to optimize neural network weights and simultaneously select the most relevant ECG features, achieving up to 95% dimensionality reduction. Experiments conducted on the UCI Arrhythmia dataset and a proprietary ECG dataset demonstrate that the GA-NN approach outperforms conventional classifiers such as SVM, k-NN, and Naïve Bayes. A key strength of the paper lies in its integration of rule extraction techniques (C4.5, RIPPER, PART), enabling the system to generate human-readable rules for clinical decision support. The analysis confirms that combining evolutionary optimization with neural networks improves both classification accuracy and model interpretability, which is critical in medical applications.

**[4] Diagnosis of heart disease using genetic algorithm based trained recurrent fuzzy neural networks, Kaan Uyar, Ahmet İlhan, 2018.**

This study uses GA based trained RFNN approach on UCI Cleveland dataset for diagnose the heart disease. The results of this study achieved a testing set (45 instances) accuracy of 97.78% and an overall accuracy of 96.63%. However, the UCI Cleveland dataset is a small, due to that reason the reliability of the model could be low. The model was not tested for overfitting. The authors also suggest the use of a larger dataset in the future research. This model is also not tested on any larger dataset so we intend to use our model on larger and diverse datasets in order to further the research.

**[5] Neural Network based Intelligent System for predicting heart disease, K. Subhadra, Vikas B, 2019.**

This paper proposes a diagnostic system for predicting heart disease. For diagnosis of heart disease 14 significant attributes are used in proposed system as per the medical literature. This paper proposed intelligent system developed using the concept of Multilayer Preception Neural Network with back propagation algorithms in order to make decisions based on multiple inputs such as patient's current and previous medical history. This research also has a small dataset that may cause the model overfitting and generalization issues. And the focus of the study was performance, accuracy and specificity but less thought is given to model interpretability. This research also doesn't collect any real-time information for model advancement unlike the 4<sup>th</sup> paper.

**[6] Heart Disease Prediction Using Deep Learning Neural Network Model, Sumit Sharma, Mahesh Parmar, 2020.**

This research compares the DNN model against several classical algorithms such as Logistic Regression, K-Nearest Neighbors, Support Vector Machine, Naïve Bayes, and Random Forest. Results show that the optimized DNN with Talos achieves the highest accuracy of around 90.78%, outperforming other methods. However, as discussed this research also uses a smaller dataset and could cause generalization problems and overfitting. The dataset used in this research also lacks feature diversity.

**[7] An Optimized Neural Network Using Genetic Algorithm for Cardiovascular Diseases, Jan Carlo T. Arroyo, Allemar Jhone P. Delima, 2022.**

To improve the accuracy of ANN in cardiovascular disease prediction, this paper employs an optimization strategy in the neural network. The parameter tuning method is achieved with the use of the genetic algorithm. In general, this study is hoped to contribute to the two significant areas of literature. First is on the disease prediction mining on how medical dataset can be processed and mined, and second, as a basis for an improved prediction model in the quest to achieve the optimal prediction accuracy. This research however has a moderate accuracy of 73.43% with the use of GA-ANN our goal is to advance on this research and achieve higher

accuracy with a larger dataset. The authors also suggest the use of more advanced Genetic Algorithms in order to improve the accuracy of the model.

**[8] A hybrid GA and PSO optimized approach for heart-disease prediction based on random forest, 2022, Mohamed G. El-Shafiey, Ahmed Hagag, El-Sayed A. El-Dahshan, Manal A. Ismail, 2022.**

This paper presents a system called GAPSO-RF that combines Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) to select the most important features for predicting heart disease. After selecting these features, a Random Forest (RF) classifier is used to make the final prediction. The system was tested on two popular datasets, Cleveland and Statlog. GA performs a global search to explore the data broadly, while PSO focuses on local search to fine-tune the solution. Using this hybrid approach, the system achieved high accuracy, with 95.6% accuracy on the Cleveland dataset and 91.4% on the Statlog dataset.

**[9] Prediction of heart disease using neural network, Snehal B. Gavande, Prof. Pramila M. Chawan, 2022.**

This paper uses the Cleveland coronary disorder dataset, which has 303 instances and 14 attributes, was used to train a model for predicting heart issues, where 80% of the data (242 instances) was trained and 20% (61 instances) was used to test. Two experiments were performed, using both a deep neural network (DNN) with more than one hidden layer and an artificial neural network (ANN) with a single or double hidden layer to compare their performance. The DNN performed better than the ANN, with its accuracy standing at 95% against the ANN's 70%, and feature selection was utilized to eliminate redundant features and provide the neural network with relevant information. The suggested work is to create a web application based on DNNs and ANNs to identify heart issues, and the intention is to design a system that offers high accuracy, and the results illustrate the efficiency of the DNN in meeting this intention.

**[10] Heart Disease Prediction Using Genetic Algorithm with Machine Learning Classifiers, Basant Abdel Menem Metwally, Nagham Elsayed Mekky, Ibrahim Mahmoud Elhenawy, 2022.**

This paper presents an optimized heart disease prediction framework that integrates Genetic Algorithms with classical machine learning classifiers to improve diagnostic performance. The GA is employed for feature selection, enabling the identification of the most influential clinical attributes before classification. The proposed system is evaluated using the Cleveland and Framingham datasets and tested with classifiers such as KNN, SVM, Decision Trees, Random Forest, and Logistic Regression. The results show that GA-optimized models significantly outperform non-optimized classifiers, achieving 100% accuracy on the Cleveland dataset and 91.8% on the Framingham dataset. The analysis highlights the effectiveness of evolutionary optimization in handling high-dimensional medical data and improving prediction reliability. This work reinforces the importance of hybrid GA–ML models in developing accurate and practical clinical decision support systems.

**[11] An intelligent system for heart disease diagnosis using regularized deep neural network, P. Rani, R. Kumar, A. Jain, 2023.**

In this paper, authors have proposed an intelligent system to predict heart disease using a Regularized Deep Neural Network (Reg-DNN). For prediction, 13 important medical features have been used. The model is trained using the Cleveland heart disease dataset. To avoid overfitting, dropout (0.3) and L2 regularization have been applied. As discussed previously the Cleveland dataset is a small so the model could have generalization and overfitting issues. And this research doesn't test the model on any other dataset either. In future, authors have planned to collect the real-time data through a user interface and assist the doctors. Further, authors have also planned to increase the accuracy of the system by searching for optimum configuration of the network by using more sophisticated algorithms like genetic algorithm.

**[12] Harnessing Deep Neural Networks for Heart Disease Prediction, Bobburi Naga Vardhana, Basati Rohitha, Gunti Anusha, Bulla Sneha Latha, Chilaka Aiswarya, Mr. R. Sudha Kishore, 2024.**

This paper uses a deep learning repository using Electrocardiogram (ECG) data, using a dataset of 334 samples with 22 corresponding clinical parameters for training a feedforward neural network with a particular architecture: an input layer



of 20 neurons using ReLU activation, three hidden layers of 64, 32, and 16 neurons, and an output layer of 2 neurons using SoftMax activation. The model is trained with categorical cross entropy loss and then trained for 15 epochs, with a precision of 98.51% when predicting heart disease, where 0 denotes no heart illness and 1 denotes heart disease, showing the efficiency of the Deep Neural Network (DNN) model in classifying heart disease correctly from ECG data.

**[13] Genetic Algorithm-based Convolutional Neural Network Feature Engineering for Optimizing Coronary Heart Disease Prediction Performance, Erwin Yudi Hidayat, Yani Parti Astuti, Ika Novita Dewi, Abu Salam, Moch. Arief Soeleman, Zainal Arifin Hasibuan, Ahmed Sabeeh Yousif, 2024.**

This work presents a comprehensive study on improving early coronary heart disease (CHD) prediction by integrating a genetic algorithm (GA) with convolutional neural network based feature selection. Using the Cleveland heart disease dataset, the authors apply information gain to select the most relevant clinical features and transform the tabular data into an image-like representation suitable for CNN processing. GA is then employed to optimize CNN hyperparameters, addressing limitations of traditional tuning methods such as grid or random search. The optimized CNN is used as a feature extractor, and its outputs are fed into conventional machine learning classifiers, including naïve Bayes, support vector machine, decision tree, logistic regression, and random forest. Experimental results demonstrate that the proposed GA-based CNN framework significantly outperforms traditional approaches, achieving up to 85% accuracy during CNN optimization and reaching 100% accuracy when combined with machine learning classifiers for both binary and multiclass CHD prediction. The study concludes that GA-driven CNN feature engineering is a powerful and reliable approach for enhancing CHD prediction performance, with strong potential for future clinical deployment and integration into AI-driven healthcare systems.



# CHAPTER 3

## SYSTEM ANALYSIS

### 3.1 Problem Statement

Heart disease prediction models today face problems like limited data, which leads to poor accuracy and overfitting. Existing methods rely heavily on costly clinical tests that take time, delaying diagnosis. Most of the existing prediction models relay of general purpose ML models to predict sensitive health scenarios. Also, even existing GA-ANN models showed lower predictive power due to small and less diverse data, low accuracy and more. Moreover, there isn't a way for user to interact with the health information and view results for themselves. Our project aims to build a more reliable system that overcomes these problems by using a larger dataset for better precision with both GA and ANN technology.

### 3.2 Proposed System.

- The proposal is to develop a project by integrating both GA hyper-parameter tuning and feature selection to train an optimized ANN model trained with a dataset of over 1,500+ samples for heart disease prediction.
- The main objective is ensuring that the most relevant input features and optimal model parameters are used during training. The optimized ANN model will be trained and validated on a suitable heart disease dataset to accurately predict the likelihood of heart disease.
- The proposal also includes development of a Web UI for users to input their information to produce predictions in real-time.

### 3.3 Requirement Analysis

The following are system requirements for development of the proposed system

#### 3.3.1 Hardware Requirements

Dedicated Graphics card with a minimum of 4 GB VRAM: Training an Artificial Neural Network (ANN) involves large-scale matrix multiplications and parallel numerical computations. A dedicated GPU accelerates these operations significantly compared to a CPU by leveraging thousands of parallel cores.

Multi-core processor (Intel Core i5 or equivalent) with a minimum of 4 cores and 8 threads with Minimum 16 GB RAM, with at least 8 GB of this available to the model.

### 3.3.2 Software Requirements

Deep Learning Framework: **TensorFlow 2.3 or higher, with Keras 2.4** or higher. The Artificial Neural Network (ANN) was implemented using Keras' **Sequential API**, enabling the construction of a multi-layer feedforward neural network with dense layers, PReLU activations, batch normalization, dropout regularization, and a sigmoid-based output layer for binary classification.

Data Manipulation: **NumPy 1.19 or higher, Pandas 1.1** or higher. NumPy and Pandas were used extensively during the data preprocessing phase. Pandas facilitated loading, cleaning, and managing the heart disease dataset, including handling missing values and organizing clinical attributes into structured data frames. NumPy supported numerical operations such as array transformations, normalization, and preparation of input features for the ANN.

Model Evaluation: **Scikit-learn 0.23** or higher. Scikit-learn was employed to support dataset splitting (training, validation, and testing) and to compute evaluation metrics such as accuracy, precision, and recall. Scikit-learn utilities also aided in normalization techniques such as Min-Max scaling and standardization .

Data Visualization: Matplotlib version 3.3 or above, Seaborn version 0.11 or above. Matplotlib and Seaborn were used to visualize data distributions and exploratory data analysis results. The paper includes multiple plots illustrating feature distributions such as age, cholesterol levels, chest pain types, heart rate, and target class balance.

Genetic Algorithm: The DEAP (Distributed Evolutionary Algorithms in Python) library was used to implement the Genetic Algorithm (GA) component of the system. DEAP enabled population initialization, fitness evaluation, selection, crossover, and mutation operations.

### 3.3.3 Functional Requirements

- The system must be have executed Genetic Algorithms to select most important feature.

- The system must have a ANN model developed with configured hidden layers with appropriate Activation function such ReLU or PreLU or any others and must be optimized to avoid issues like overfitting.
- The ANN model be precisely trained and validated for accuracy, and all the factors of a confusion matrix along with Hyper-parameters Optimized from GA. Only the best performing model must be saved.
- The system must a Flask based Web interface for user to enter information, read the information and display the appropriate prediction.

# CHAPTER 4

## DESIGN

### 4.1 Methodology

One of the major limitations of existing models is their reliance on limited and often non-representative datasets, which can hinder prediction accuracy and generalizability. The use of larger dataset is highly recommended pros. Furthermore, the highest accuracy that was previously achieved from such a model was 73.93% while we aim to optimize it further and aim for high 80 – 90% accuracy. In addition to the technical enhancements, the proposal includes the development of a user-friendly web-based interface. This Web UI will not only allow users to input data and receive real-time predictions but also serve as a platform for continuous data collection. The new data will be utilized for ongoing model improvement although this may require user trust , ensuring that the system evolves and adapts over time to maintain high predictive performance.

#### Dataset:

This dataset used to Train the model contains over 1,500 samples with 13 features as opposed to Cleveland dataset used in previous studies which has 202-300 samples with 11 features. With this dataset we hope to achieve greater accuracy then that of 80%.

Table 1: Dataset Overview

| Attributes | Description                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------|
| Age        | Patient Age                                                                                              |
| Sex        | 0:Female, 1:Male                                                                                         |
| Cp         | Type of Chest Pain:<br>1. typical Angina<br>2. atypical angina<br>3. non-anginal pain<br>4. asymptomatic |
| Trestbps   | Resting Blood sugar (in mm Hg on admission to hospital)                                                  |
| Chol       | Serum cholesterols in mg/dl                                                                              |
| Fbs        | Fasting blood suger > 120 mg/dl.(1=true;                                                                 |

|          |                                                                                                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | 0=false)                                                                                                                                                                                                                                     |
| Restecg  | Resting electrocardiographic results ( 0 = normal; 1 = having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of $> 0.05$ mV); 2 = showing probable or definite left ventricular hypertrophy by Estes' criteria ) |
| Thalach  | Maximum heart rates Achieved.                                                                                                                                                                                                                |
| Exang    | Exercise induced angina.(1=yes;0=no)                                                                                                                                                                                                         |
| Old Peak | ST depression induced by exercise relative to rest.                                                                                                                                                                                          |
| Slope    | Slope or peak exercise ST Segment <ol style="list-style-type: none"><li>1. upsloping</li><li>2. flat</li><li>3. downsloping</li></ol>                                                                                                        |
| Ca       | number of major vessels (0-3) colored by flourosop.                                                                                                                                                                                          |
| Thal     | Thalassemia<br>(0 = error; 1 = normal; 2 = fixed defect; 3 = reversible defect)                                                                                                                                                              |
| Target   | Target Attribute:<br>0:Yes;<br>1:No:                                                                                                                                                                                                         |

There is notable improvement in accuracy of the ANN model when compared to the previous results. The following will be the methodology of the project and the processes performed to reach that improvement. If we had to say the improvement observed here is approximately 12%.

Thal: Thalassemia feature is removed in the final iteration of the model train and testing as Thalassemia are a group of inherited blood disorders that manifest as the production of reduced hemoglobin the presence of this disorder will lead to

heart disease eventually, which could make the model skewed toward patient's who do not have this disorder.

#### **4.1.1 Data Preprocessing**

Preprocessing is a significant phase of developing an Artificial Neural Network (ANN) model. The main operations of data processing are defined by the following. Missing Values Handling: The data is checked for missing values, which have a strong effect on the performance of the ANN model. Missing values are either assigned with mean, median, or mode imputation, or the rows with missing values are dropped, based on the extent and distribution of missing data.

Data Normalization: The attributes in the data set are normalized so that they are of the same scale. This is generally achieved with methods like Min-Max Scaler or Standard Scaler, which bring the data to a particular range (say between 0 and 1) or mean = 0 and standard deviation = 1, respectively.

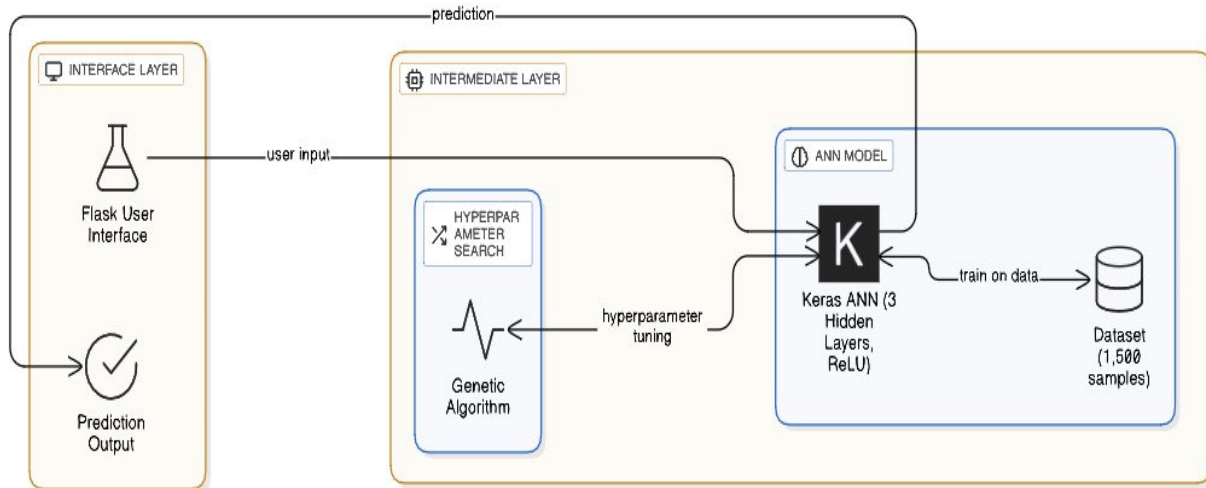
Data Split: The preprocessed dataset is further divided into three sets: training (70% of the data), validation (15% of the data), and testing (15% of the data). The training dataset is utilized to train the ANN model, the validation dataset is utilized to assess the model performance during training and optimize hyperparameters, and the testing dataset is utilized to assess the final model performance.

#### **4.1.2 Model Architecture Design**

This section represents the proposed architecture which follows a layered and modular design as displayed in Fig 1 that enables efficient prediction through a seamless flow of data from user input to final output. It integrates user interaction, intelligent processing, and model learning into a unified pipeline, ensuring flexibility, scalability, and accuracy.

A. Interface Layer:

Flask User Interface: As displayed in Fig 1, This is the front-end interface where a user can provide inputs (features of the dataset). After the user submits input data, the system processes it through the model and returns the predicted result here.



**Fig 1: System Architecture**

## B. Intermediate Layer

**Hyper-parameter Search:** This component uses a Genetic Algorithm (GA) to optimize the hyper-parameters of the ANN. Hyper-parameters include values like learning rate, number of neurons, number of hidden layers, activation functions, batch size, etc.

**Genetic Algorithm:** Works by simulating the process of natural selection (selection, crossover, mutation) to search for the best set of hyper-parameters. Takes user input and performs hyperparameter tuning. The optimized hyper-parameters are then passed to the ANN model for training.

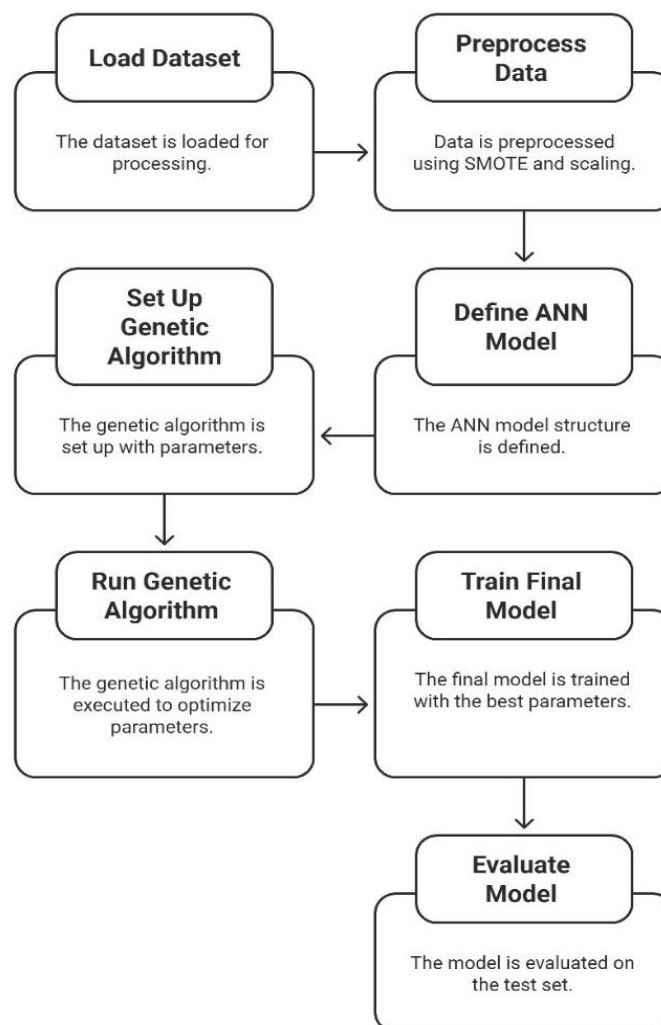
**Keras based ANN Model:** Multiple Dense layers (149, 112, and 50 neurons, respectively) using PReLU activation. Each Dense layer is followed by Batch Normalization and Dropout for regularization. The output layer uses a single neuron (for binary classification). The model summary indicates approximately 29,431 total parameters, most of which are trainable.

### 4.1.3 Model Compilation and Training:

**Compilation:** This model is compiled with the Adam optimizer and binary cross-entropy loss function. The Adam optimizer is a widely used stochastic gradient descent optimizer that learns the learning rate for each parameter separately depending on the magnitude of the gradient. Binary cross-entropy loss is

appropriate for binary classification tasks, in which the objective is to predict a probability of membership in one of two classes.

**Training:** The model is trained on the training set, and the validation set is utilized to monitor the performance of the model while training. The model is trained for a certain number of epochs, which is a hyper-parameter that requires tuning. The learning rate and batch size are also tuned to improve the training process. The model's performance on the validation set is monitored during training, and the model is saved at the epoch with the best performance.



**Fig 2. Methodology and Workflow**



#### 4.1.4 Feature Selection using Genetic Algorithm (GA):

Feature selection is a crucial step in building an accurate ANN model. A Genetic Algorithm (GA) is used to optimize the feature selection process:

**GA Overview:** A GA is a meta-heuristic based on the natural selection process. It produces a population of potential solutions, and each solution is a set of features. The fitness of every solution is calculated by the ANN model, and solutions with the greatest fitness are chosen for reproduction to generate new offspring.

**GA Parameters:** GA parameters like population size, number of generations, crossover rate, and mutation rate are adjusted to optimize the feature selection process. The GA is executed for a number of generations so that it can converge to an optimum solution. The optimum solution is the set of features that yields the optimum performance of the ANN model.

#### 4.1.5 Model Validation and Evaluation

As described in Fig 2 this section focuses of Validation and Evaluation, The model's performance is also validated using the validation set, ensuring that the model is generalizable and not over-fitting to the training data. The model's performance on the validation set is compared to its performance on the testing set to ensure that the model is not overfitting. With all this we can report the accuracy of 88.3% at the highest with the accuracy being between 80~90% constantly. Further information on evaluation in Testing and results section.

Table 2. Previous research results

| Prediction Model | Accuracy |
|------------------|----------|
| ANN              | 68.35%   |
| GA+ANN           | 73.43%   |

Table 3. Our Findings

|                 |        |
|-----------------|--------|
| ANN             | 77.77% |
| GA + ANN(PReLU) | 88.33% |

Table 4. Evaluation Matrics of the Best model

| Metric               | Value  |
|----------------------|--------|
| Accuracy             | 88.33% |
| Precision            | 89.82% |
| Recall (Sensitivity) | 86.33% |
| F1-Score             | 88.04% |
| AUC-ROC              | 0.91   |

# CHAPTER 5

## IMPLEMENTATION

### 5.1 Implementation modules

#### Keras ANN model:

The Artificial Neural Network (ANN) model was implemented using the Keras API on top of the TensorFlow framework. Keras was selected due to its modular design, ease of experimentation, and strong support for rapid prototyping of deep learning models. The ANN architecture consists of an input layer, three hidden layers, and an output layer. Each hidden layer uses the **Parametric Rectified Linear Unit (PReLU)** activation function. PReLU was chosen because it allows the negative slope of the activation function to be learned during training, which helps improve model convergence and reduces the risk of dying neurons compared to standard ReLU.

#### GA for Hyper-parameter Tuning:

Genetic Algorithm hyper-parameter tuning is a meta-heuristic way to automatically search for the best set of model hyper-parameters (like number of neurons, learning rate, dropout, regularization, batch size, etc.). The following section describes each step of the GA implementation.

**Representation (Chromosome / Individual):** In this Genetic Algorithm (GA) implementation, each individual, or chromosome, is represented as a list that combines two essential components: a binary feature mask and a sequence of numeric hyper-parameters. The binary feature mask has a length equal to the number of input features, where each bit corresponds to a feature and is set to either 0 or 1, indicating its inclusion or exclusion. The sequence of numeric hyper-parameters follows the order defined by **hyprparameter\_ranges**. Utilizing the **creator.Individual** type, each individual inherently carries a fitness value, which is crucial for evaluating its quality during the evolutionary process.

**Initialization:** The initialization of individuals is performed by the **smart\_individual** function, which takes **ranked\_features** and other parameters into account. This function constructs an initial individual by randomly selecting a subset of top-ranked features (typically between 6 to 10) and setting their corresponding bits in the feature

mask to 1. It then appends a vector of random hyper-parameter values, generated by **random\_hyprparameters**, to the individual. The **toolbox.individual** and **toolbox.population** functions are used to create the initial population of a specified size (**population\_size**), comprising individuals generated by **smart\_individual**.

**Fitness Function / Evaluation:** The fitness of an individual is evaluated by the evaluate function, which takes an individual, the input shape, and the current generation as inputs. It first extracts the feature mask and hyper-parameters from the individual. Using these, it constructs training and test subsets of the data, selecting only the features indicated by the feature mask. A neural network model is then created, compiled, and trained with the specified hyper-parameters and feature set through the **create\_ann\_model** function. The model predicts on the test set, and the accuracy of these predictions serves as the fitness of the individual. Additionally, the selected features, hyper-parameters, and accuracy are logged to a CSV file for record-keeping.

### **User Interface(UI/UX):**

A Responsive Flask Web Application to Take User inputs in text and selection fields to generate prediction from pretrained model. Fig 5 shows the final design of the Web UI. Although simple it is functional and the simplicity makes it ease to use and clear.

## **5.2 Development Environment**

**Jupyter Notebook:** Jupyter Notebook was used as the primary development environment for data exploration, preprocessing, model training, and evaluation. Its interactive nature allowed for efficient experimentation, visualization of results, and comparative analysis of different model configurations. The notebook environment was particularly useful for monitoring training behavior and analyzing performance metrics across different activation functions and hyper-parameter settings.

**Neovim:** Neovim code editor was used for development and testing of the web application and frontend of the project. Neovim is a highly extensible and customizable text editor that is a fork of the popular Vim editor. It is designed to be a more modern and community-driven alternative to Vim, with a focus on extensibility, usability, and performance.

**Git:** Git was used for the version control of the project during its development. Git is a distributed version control system that tracks changes to code over time, allowing developers to create a history of changes, revert to previous versions, and compare changes between different versions.

**GitHub:** Github platform was used to store code and pre trained model files during the development of the project.

# CHAPTER 6

## TESTING AND RESULTS

This chapter presents a comprehensive evaluation of the proposed model through systematic testing and analysis to assess its performance and stability. Multiple experiments were conducted to identify the most suitable configuration for accurate heart disease prediction, with a primary focus on comparing different activation functions under controlled conditions. Each model variant was trained using a consistent architecture and number of epochs to ensure a fair comparison. The evaluation was carried out using quantitative performance metrics such as accuracy, along with detailed confusion matrix analysis to examine class-wise prediction behavior.

### 6.1 Activation functions testing

The following test were performed in order to determine the best performing activation function for this particular use case.

The models were tested with activation functions including ReLU, LeakyReLU, PReLU, ELU, SELU, Tanh. While the performance of these models was relatively similar the **PReLU** activated model resulted in the highest accuracy and consistency.

Table 5: Model Comparison

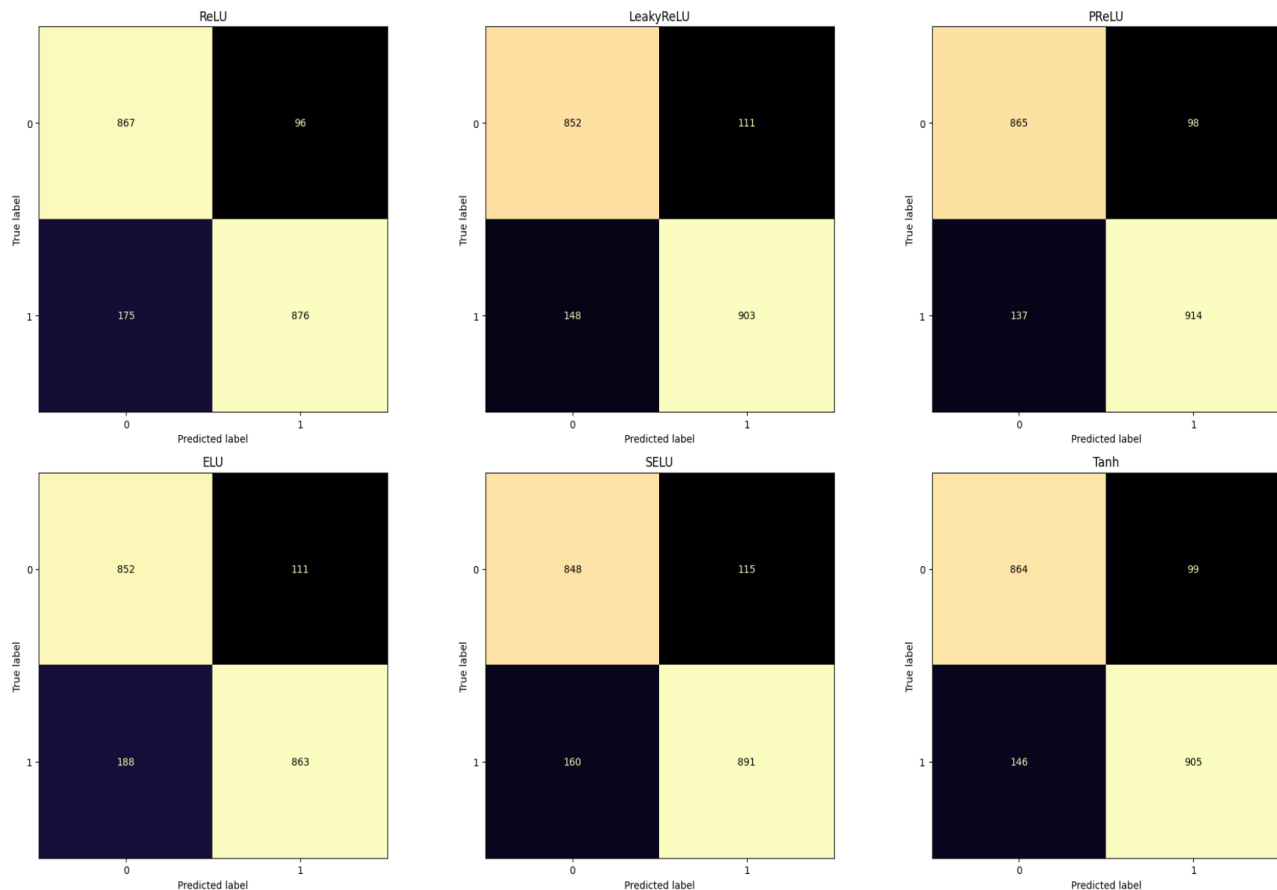
| Activation      | Neurons<br>1 <sup>st</sup> H layer | Neurons<br>2 <sup>nd</sup> Layer | Neurons<br>3 <sup>rd</sup> Layer | Epochs | Accuracy |
|-----------------|------------------------------------|----------------------------------|----------------------------------|--------|----------|
| ReLU_model      | 137                                | 76                               | 53                               | 30     | 86.54%   |
| LeakyReLU_model | 163                                | 86                               | 40                               | 30     | 87.14%   |
| PReLU_model     | 149                                | 112                              | 50                               | 30     | 88.33%   |
| ELU_model       | 174                                | 92                               | 45                               | 30     | 85.15%   |
| SELU_model      | 140                                | 94                               | 33                               | 30     | 86.35%   |
| Tanh_model      | 185                                | 80                               | 54                               | 30     | 87.84%   |

### 6.2 Confusion Matrix Analysis

The Accuracy isn't enough to formulate the final judgement on which model performs the best Therefore confusion matrix is used to understand **how the model performs on each class**, rather than just giving a single accuracy number.

The confusion matrix Fig 3 helps determine how consistently the model correctly predicts the heart disease. Hence there are 4 categories to the confusion matrix. The most important is True Positive shows if model correctly predicts the disease and True Negative shows if model correctly predicts the absence of the disease. Every Activation model was tested with confusion matrix.

The following fig 3 shows the best performing model that is PReLU model. The figure shows 865 True Positives, 914 True Negative, 98 False Positive, 137 False Negative.



**Fig 3: Confusion Matrix**

**ScreenShots:**

**Heart Disease Prediction**

Age (years):  Sex:

Chest pain type:

Resting Blood Pressure (mm Hg):

Serum Cholesterol (mg/dl):

Fasting Blood Sugar > 120 mg/dl:

Resting ECG Results:

Max Heart Rate Achieved:  Exercise Induced Angina:

Oldpeak (ST depression):  Slope of ST segment:

Number of Major Vessels (0-3):

**Heart Disease Prediction**

Age (years): 43 Sex: 1 - Male

Chest pain type: 0 - Typical angina

Resting Blood Pressure (mm Hg): 140

Serum Cholesterol (mg/dl): 201

Fasting Blood Sugar > 120 mg/dl: 1 - True

Resting ECG Results: 0 - Normal

Max Heart Rate Achieved: 145 Exercise Induced Angina: 1 - Yes

Oldpeak (ST depression): 3.1 Slope of ST segment: 0 - Upsloping

Number of Major Vessels (0-3): 0

Low Likelihood of Heart Disease.

**Fig 4: Output Web UI**



# CONCLUSION

The developed neural network model for heart disease prediction was trained and evaluated on a dataset of 1,510 instances, each representing patient clinical features related to cardiovascular health. The model achieved an accuracy of **88.33%** on the held-out test set. This level of accuracy suggests the neural network was able to effectively capture patterns and relationships in the clinical features that are indicative of heart disease presence. We also believe the model can be improved with more advanced technology and our suggestion for the future research is to consult actual medical professionals as an on field research to better understand the cardiovascular state over progressive stages of treatment.

In summary, the neural network approach yielded promising results and represents a viable step toward automated, data-driven heart disease prediction, but careful consideration of medical context, model interpretability, and diagnostic implications remains essential.

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